www.tntextbooks.in

Solution:



When the block slides, the force acting on the block is kinetic friction which is equal to $f_k = \mu_k mg$

From Newton's second law $ma = -\mu_k mg$

The negative sign implies that force acts on the opposite direction of motion.

The acceleration of the block while sliding $a = -\mu_k g$.

The negative sign implies that the acceleration is in opposite direction of the velocity.

Note that the acceleration depends only on g and the coefficient of kinetic friction μ_k

We can apply the following kinematic equation

$$v = u + at$$

The final velocity is zero.

$$0 = u - \mu_k gt$$

$$t = \frac{u}{\mu_k g}$$

7. Three blocks of masses 10 kg, 7 kg and 2 kg are placed in contact with each other on a frictionless table. A force of 50 N is applied on the heaviest mass. What is the acceleration of the system?

Solution:



We know that

$$a = \left[\frac{F}{m_1 + m_2 + m_3}\right] = \frac{50N}{10kg + 7kg + 2kg}$$

$$= \frac{50}{19} = 2.63\, ms^{-2}$$

8. The coefficient of friction between a block and plane is $\frac{1}{\sqrt{3}}$. If the inclination of the plane gradually increases, at what angle will the object begin to slide?

Since the coefficient of friction is $\frac{1}{\sqrt{3}}$

$$\text{Tan}\,\Theta = \frac{1}{\sqrt{3}} \Rightarrow \Theta = 30°$$

9. Find the maximum speed at which a car can turn round a curve of 36 m radius on a level road. Given the coefficient of friction between the tyre and the road is 0.53.

Radius of the curve r = 36 m

Coefficient of friction μ = 0.53

Acceleration due to gravity g = 10 ms^{-1}

$$v_{max} = \sqrt{\mu rg} = \sqrt{0.53 \times 36 \times 10}$$

$$= 13.81\, ms^{-1}$$

10. Calculate the centripetal acceleration of the Earth which orbits around the